

## Assignment Instructions: Development of a Recommender System

### Deadline

On or before 23rd February 2026 – 11:59 PM

Late submissions will not be allowed.

### Objective

Design and implement a working recommender system for a real-world application using a suitable recommendation algorithm. You must demonstrate your system, evaluate its performance, and explain your design decisions.

### 1. Choose an Application Domain

Select ONE domain (or propose your own):

- Movie / TV recommendation
- E-commerce products
- Music playlist generation
- News article recommendation
- Books / academic paper recommendation
- Restaurants / travel recommendation
- Online learning resources

Your application must include user-item or user-user or item-item interaction data.

### 2. Dataset Requirements

You may use public datasets (MovieLens, Amazon Reviews, Kaggle, etc.) OR create a realistic simulated dataset.

Minimum expectations:

- At least 500 interactions
- At least 50 users
- At least 50 items

You must explain users, items, interaction signal, and dataset sparsity level.

### 3. Algorithm Requirement

- Implement at least ONE of the following:

#### Content-Based Filtering:

- TF-IDF + cosine similarity
- Embedding-based similarity

#### Collaborative Filtering:

- User-based CF
- Item-based CF
- Matrix factorization (SVD, ALS, etc.)

- **Knowledge based approach**

- **AI based approach**

- **Hybrid Recommender (Bonus):**

- Combination of content + collaborative and another other following any hybrid recommendation framework.

You must justify your algorithm choice for the chosen domain

#### 4. Mandatory Development Steps

##### **Step 1 – Problem Definition**

- What is being recommended?
- Who are the users?
- Objective (Top-N or Rating Prediction?)

##### **Step 2 – Data Preparation**

- Cleaning
- Train/test split
- Feature preprocessing

##### **Step 3 – Model Development**

- Implement algorithm
- Train model
- Generate recommendations

##### **Step 4 – Evaluation**

Use at least TWO metrics.

Top-N: Precision@K, Recall@K, NDCG@K, Hit Rate@K

Rating Prediction: RMSE, MAE

**Compare with a baseline (Most Popular / Random / Global Average).**

##### **Step 5 – Demonstration**

- Show recommendations for at least 3 users
- Provide brief explanation
- Show performance comparison table

## 5. Deliverables (What to Submit)

Submit a single ZIP file named: RegNo\_Name\_RecommenderSystem.zip

Include:

- Python Code (.ipynb or .py)
- Dataset or dataset link
- README.txt (how to run, libraries, expected output)
- 3-Minute Video Presentation (MP4 format)

## Video Presentation Requirements (3 Minutes – Mandatory)

The video must include:

1. Problem/domain introduction
2. Dataset description
3. Algorithm used and justification
4. Evaluation results
5. Live demonstration

Maximum length: 3 minutes

Clear audio

Professional presentation style

## 6. Marking Scheme (100 Marks)

- Problem definition & domain suitability – 10
- Dataset preparation & analysis – 20
- Algorithm implementation & justification – 25
- Evaluation & baseline comparison – 25
- Code clarity & reproducibility – 10
- Video presentation quality & clarity – 10

Bonus (up to +10): Hybrid model, advanced evaluation, cold-start handling, time-aware recommendation.

## 7. Academic Integrity

Individual work

Plagiarism will be penalized.

AI tools may be used for assistance, but students must understand their work.